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Detecting safety helmet wearing on construction sites with bounding-box regression and deep transfer learning

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Abstract

Detecting safety helmet wearing in surveillance videos is an essential task for safety management, compliance with regulations, and reducing the death rate from construction industry accidents. However, it is much challenged by some factors like interocclusion, scale variances, perspective distortion, small object detection, and the carrier recognition of safety helmet. Traditional image-based methods suffer from them. This article proposes a new methodology for detecting safety helmet wearing, which makes use of convolutional neural network-based face detection and bounding-box regression for safety helmet detection. On the one hand, the method can help to recognize the carrier of the safety helmet and detect a multiscale and small safety helmet. On the other hand, deep transfer learning based on DenseNet is introduced and applied using two different strategies, namely, object feature extractor and fine-tuning for safety helmet recognition. To further improve the recognition accuracy, the network model with two peer DenseNet networks is trained by mutual distillation. Extensive analysis and experiments show that the novel methodology has considerable advantages in detecting safety helmet wearing compared to other state-of-the-art models. The proposed model has achieved 96.2% recall, 96.2% precision, and 94.47% average detection accuracy. These results, precision-recall (PR) curve, and receiver operating characteristic (ROC) curve demonstrate the feasibility of the new model.

1 | INTRODUCTION

With the development of civil and infrastructure engineering, the issues related to workers on construction sites are getting more and more attention. The research on workers' activities can help to understand workers' time allocation (Luo, Li, Yang, & Yu, 2019), and the study on the work–rest schedule for workers in hot weather is to maximize the total production time and avoid the adverse effect of heat stress (W. Li & Wang, 2017). In addition, identifying onsite objects and understanding their motion characteristics in construction work zones are helpful for safe equipment operation and worker safety by identifying valuable information on these objects (Caldas & Kim, 2009; S. Chi & Caldas, 2011). Currently, the research on safety issues is one of the main challenges for workers on construction sites, and the automatic detection for onsite safety, such as the use of personal protective equipment (PPE), especially safety helmet, has received continuous attention.

Safety helmet belongs to one of the crucial PPEs of construction engineering, and it is an effective PPE to protect workers from being struck by falling objects (Rubaiyat et al., 2016; Suderman, Hoover, Ching, & Scher, 2014). Moreover, wearing a safety helmet on a construction site is a statutory requirement all over the world (Cabahug, 2014). However, some onsite workers would not like to obey the regulation to wear a safety helmet because of discomfort. Its automatic

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detection is hence vital for onsite workers' safety, and it is significant for improving safety management level to detect and track the equipment automatically. Instead, it is impractical and expensive for traditional manual detection based on labor-intensive methods to acquire these data from a construction site or video recording.

In the past decade, some technologies have been researched and developed for detecting safety helmets. For example, three-axis accelerometer sensors were utilized for identifying the safety helmet is being adequately worn or not worn (Kim, Wang, Min, & Lee, 2018). Other sensors such as chinstrap sensors, altitude sensing data, and radio frequency identification (RFID) were also employed for safety helmet detection (Barro-Torres, Fernández-Caramés, Pérez-Iglesias, & Escudero, 2012; Kelm et al., 2013; Zhang, Yan, Li, & Jin, 2019). However, these methods have no way to identify whether the equipment is held, worn, or placed on the other places. Dong, He, Li, and Yin (2015) used a pressure sensor to identify whether the safety helmet was being properly worn and Bluetooth sensor to transmit information. However, the Bluetooth device needs to be charged after periodic use, and its communication range is limited. Furthermore, the use of these sensors increases the up–front investment. Wearing a safety helmet with a sensor can be regarded as intrusive to a worker, and many workers are reluctant to wear such equipment due to privacy and health issues. Therefore, the issue of recognizing the carrier of a safety helmet is challenging.

Currently, the most commonly deployed sensor for automatic detection on a construction site is the imaging sensor using visible light. Many algorithms have been proposed to solve the problem of safety helmet detection and similar applications in videos and images (W. Fang, Ding, Luo, & Love, 2018; Park, Elsafty, & Zhu, 2015; Rubaiyat et al., 2016; Seong, Son, & Kim, 2018; Shrestha, Sherestha, Bajracharya, & Yfantis, 2015). However, the imaging sensor-based method is hugely challenged by some factors such as interocclusion, scale variation, perspective distortion, and small object detection. Traditional safety helmet detection methods based on computer vision and image recognition include histogram of oriented gradient (HOG)-based approaches (Park et al., 2015; Rubaiyat et al., 2016; Zhu, Park, & Elsafty, 2015), color-based approaches (K. Li, Zhao, Bian, & Tan, 2018), a hybrid descriptor-based approach (Silva, Aires, & Veras, 2014; Silva et al., 2013; Wu & Zhao, 2018), and other methods (Waranusast, Bundon, Timtong, Tangnoi, & Pattanathaburt, 2013). These methods for safety helmet detection suffer from those factors above, which are shown in Figure 1 (column 1).

For example, there are different degrees of interocclusion for safety helmets and individuals in Figure 1 (column 1). The traditional image-based methods (K. Li et al., 2018; Park et al., 2015; Rubaiyat et al., 2016; Silva et al., 2014; Waranusast et al., 2013; Zhu et al., 2015) seldom discuss occlusion issue, and they are not suitable for the crowd scene shown in Figure 1 (row 1 and column 1). Q. Fang et al. (2018) first researched the issue in-depth, but the recall rate is only 64.5% when part of a head visible. Therefore, the occlusion issue is very challenging for object detection (Loy et al., 2018).

Moreover, there is a significant perspective distortion for safety helmets in Figure 1 (row 2 and column 1) that are closer to the surveillance camera and appear indeed larger than those that are further away from the surveillance camera. Meanwhile, the scale differences for all safety helmets that are coupled by perspective distortion in each image are evident in Figure 1 (column 1). The scale variation between the smallest helmet size and the largest helmet size is over 20 times, and it leads to an extreme challenge to an object detection model (Loy et al., 2018). What is far worse is that the detected object is smaller than 1% of the whole image area, so it is very challenging to detect the small object (Cai & Vasconcelos, 2018). To avoid detecting a small object, most methods for detecting safety helmets first made use of pedestrian detection (Q. Fang et al., 2018; K. Li et al., 2018; Rubaiyat et al., 2016; Zhu et al., 2015), and then identified safety helmets. However, the pedestrian detection–based method suffers from occlusion issues shown in Figure 1. The images in the first column are raw images, including 50 and 11 workers, respectively. The pedestrian detection-based method (Chen et al., 2019) detects 27 workers from the first raw image, and the face detection-based method (J. Li et al., 2019) detects 42 workers from the first raw image. The two methods achieve the same detection numbers from the second raw image. The testing results shown by the red numbers on images indicate that the pedestrian detection method is particularly vulnerable to occlusion in crowd scenes. On the other hand, the pedestrian detection method can detect nonhelmet-use workers but not recognize who they are, and the evidence for these nonhelmet-use workers is also essential. Thus, the carrier recognition of a safety helmet is still a challenging issue.

In addition, there is previously no public and open data set for developing and evaluating safety helmet detection algorithms under various challenging issues. In practice, the challenging issues for safety helmet detection usually influence each other, and we have no way to test and evaluate these algorithms using uniform measure on a public data set.

With the advance of many advanced methods, especially tiny face detection, a new method is proposed using face bounding-box regression and deep transfer learning, which can make use of face bounding box to calculate safety helmet bounding box and then identify it in the candidate region using the convolutional neural network with deep transfer learning. In this article, we aim to adopt a face detection–based method to yield an innovative solution for safety helmet detection with interocclusion, scale variances, perspective distortion, and small object detection in various construction sites. The main contributions in this work are as follows: (a) We propose a new methodology for detecting safety



FIGURE 1 The testing results for occlusion issue

helmet wearing in various construction sites where interocclusion, scale variances, perspective distortion, and small objects occurred. (b) More specifically, we develop a face bounding-box regression algorithm with the detection of the face of various sizes to determine the candidate region of a safety helmet. (c) We make use of two different strategies for deep transfer learning based on DenseNet (Huang, Liu, van der Maaten, & Weinberger, 2017), namely, fine-tuning and mutual distillation methods for improving safety helmet recognition. (d) We present extensive analysis and experiments based on the DataFountain's public data set and verify state-of-the-art performance.

The rest of the article is organized as follows. We first summarize the literature review of safety helmet detection and tiny face detection methods in Section 2. Section 3 describes our proposed methodology in detail. The experiments for our models are implemented and explained in Sections 4 and 5. Testing and comparative studies are illustrated and analyzed in Section 6, and Section 7 concludes the study.

2 | LITERATURE REVIEW

2.1 | Safety helmet detection

Vision-based object detection has been extensively studied in recent years, and many methods for detecting safety helmet used computer vision and image recognition technologies. Waranusast et al. (2013) used KNN (K-nearest neighbors) classifier to identify the color and shape information extracted from moving objects and detect helmet, and the detection accuracy is 74%. Dahiya, Singh, and Mohan (2016) compared the three feature descriptors that were widely used for helmet detection, HOG, scale-invariant feature transform (SIFT), and local binary patterns (LBP) with support vector machine (SVM) classifier, and the results showed that HOG descriptor has the best performance. However, the HOG-based method suffers from helmet-like objects with similar edge features such as a head and a turn in the road, and these objects are misidentified as a helmet. Therefore, it is difficult for only one descriptor to identify helmets.

Silva et al. (2013) utilized a hybrid descriptor including HOG, LBP, and circle Hough transform (CHT) to extract edge, texture, and geometric features of helmet, and the reported accuracy is 94.23%. In addition, Silva et al. (2014) tried to combine wavelet transform (WT), HOG, and LBP features, and the method returns a 91.37% accuracy rate. Rubaiyat et al. (2016) combined the color and CHT to extract features, and the proposed method has an 81.0% accuracy rate. Wu et al. (2018) made use of LBP, Hu moment invariants (HMI), and color histograms (CH) features, and the average accuracy rate is 90.3%.

However, the above methods usually are based on moving objects and pedestrian detection, and the occlusion that is shown in Figure 1 is a severe problem. Therefore, these methods chose the test samples from some simple scenarios. Meanwhile, these methods commonly detected safety helmets in near-field videos, and they are unable to talk about scale variances, perspective distortion, and small object detection issues. Moreover, using moving objects and pedestrian detection, these methods have no way to identify the carrier of a safety helmet. Finally, since there is no public data set with various challenging issues, it is tough to compare and assess these algorithms.

In addition, little work has been done to study the face detection–based method. Du, Shehata, and Badawy (2011) and Shrestha et al. (2015) used a traditional computer vision method to detect the face and helmet. However, their algorithms required a high-resolution closed-circuit television (CCTV) system, and workers should face to the camera.

Moreover, the two systems were not verified on various construction sites.

In recent years, the deep learning-based methods have been widely used in object detection (Gao & Mosalam, 2018; Molina-Cabello, Luque-Baena, López-Rubio, & Thurnhofer-Hemsi, 2018; Rafiei, & Adeli, 2017c; Vera-Olmos, Pardo, Melero, & Malpica, 2019) and other engineering fields (Nabian & Meidani, 2018; Rafiei, & Adeli, 2017a, 2018; Shen, Xiong, Xue, & Bian, 2019; Wang & Bai, 2018), so these methods have motivated deep learning-based safety helmet wearing detection. Q. Fang et al. (2018) proposed an automatic non-hardhat-use (NHU) detection technology that used an improved faster region-based convolutional neural networks (Faster R-CNN) method (Ren, He, Girshick, & Sun, 2017). They collected 81,000 images with various challenging issues such as scale, occlusion, and small object detection. The NHU workers were labeled as the ground truth during the training phase, and they were detected during the test phase. The rest were identified as the background. The proposed method has an accuracy rate between 90.1% and 98.4% that depends on the impact of different situations. Since the method only detects the NHU worker, it has no way to identify safety helmet colors that represent different roles on sites. Besides, the method based on pedestrian detection is unable to recognize the carrier of safety helmet, and it suffers from occlusion issues shown in Figure 1. Furthermore, the data set is not open to access.

It was evident from the above discussion that more and more research work has gone into safety helmet wearing detection in recent years, and most of these methods had some limitations for safety helmet wearing detection, especially in dealing with interocclusion, scale variances, perspective distortion, small object detection, and the carrier recognition of safety helmet. Aiming at these issues, we need to present a new algorithm and improve performance. This article proposes a new automatic safety helmet wearing detection method based on deep transfer learning and face bounding-box regression with tiny face detection to explore the limitations. We verify the advantages of our method on the public data set based on Datafountain's data set, which is publicly available (https://github.com/LP940708/Safety-Helmet-Classifier-Dataset).

2.2 | Tiny face detection

Earlier methods for object detection based on traditional computer vision such as HOG and LBP have been challenged due to the difficulties in learning and training procedures (Szegedy, Toshev, & Erhan, 2013). Krizhevsky, Sutskever, and Hinton (2012) trained a deep convolutional neural network for ImageNet classification, and their work promoted the deep learning method, which has been used in many science fields, especially in object detection. On one hand, Faster R-CNN (Ren et al., 2017), you only look once (YOLO) (Redmon, Divvala, Girshick, & Farhadi, 2016), and single shot detector (SSD) (Liu et al., 2016) are widely used for object detection at present. However, there are no public and extensive data sets for training a good detection model of safety helmet, so it is impractical for safety helmet detection to make use of these models directly. Instead, face detection is one of the most successful areas of artificial intelligence research, and one of the main reasons is to accumulate various challenging data sets for training an advanced face detection model. Inspired by the face detection models, a new algorithm is proposed based on the face bounding-box regression, which can generate a safety helmet bounding box through the result of face detection. On the other hand, the above methods belong to universal object detection models, and so they are widely used for pedestrian detection (Q. Fang et al., 2018), face detection (C. Chi et al., 2019; C. Z. Zhang, Xu, & Tu, 2018), and other fields. However, there are some issues for face detection that existed in feature learning, loss design, and anchor matching.

Early works used CNN as a sliding detector on image pyramids to learn features, which leads to low speed and inefficient memory. Therefore, these detectors had to extract features with a single scale. Besides, the region proposal network (Ren et al., 2017) and region of interest pooling are applied to acquire features in each region. However, new research shows that multiscale-based features are better for detecting tiny objects. For example, feature pyramid networks with top–down structure (Lin et al., 2017) made use of all scales with high-level semantic information. PyramidBox based on feature pyramid networks (Tang, Du, He, & Liu, 2018) dramatically improves object detection. The dual shot face detector (DSFD) (J. Li et al., 2019) considers layer information and context relationship between anchors to enhance the features. Moreover, DSFD adopts progressive anchor loss (PAL) for improving the conventional loss functions, and it considers random sampling in data augmentation and makes anchor matching better. Therefore, we use DSFD for face detection on various construction sites.

3 | METHODOLOGY

The section summarizes the training and testing process of a safety helmet detection model shown in Figure 2. The flowchart illustrates how to make use of face detection, training set, validation set, and testing set to acquire a face-to-helmet regression model (FHRM), a deep transfer learning-based helmet detection model (DTLHDM), and a complete safety helmet detector, which is used to infer the safety helmet wearing detection in a surveillance image directly. Raw images of various construction sites with as many challenging issues as possible are taken from different surveillance

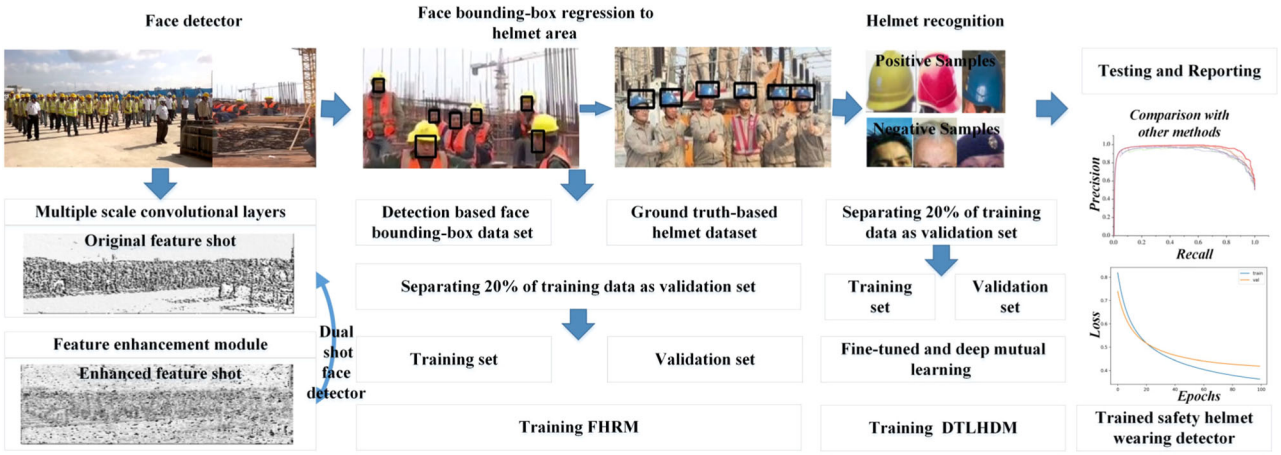


FIGURE 2 Flowchart for safety helmet detection

camera settings, and these images are used to train a safety helmet wearing detector. The definition of a safety helmet in this article is that a pedestrian with the safety helmet should be identifiable in pedestrian images via our naked eye.

To train the FHRM, we randomly selected safety helmet samples in the data set with ground-truth labels to detect the corresponding faces and acquire their bounding boxes. The FHRM training data were composed of these face data with bounding boxes and helmet data with ground-truth data. We separated 20% of the training data as a validation set, and the rest of the training data were fed into the FHRM to build a regressor for outputting the bounding box of a safety helmet in the validation data set. When the validation data set validates the regressor, safety helmets in the validation data set acquired by the regression-based method and ground-truth-based method, respectively, were analyzed and compared to get a report of regression precision. Meanwhile, to train the DTLHDM, positive and negative samples were randomly chosen from the corresponding data sets, and they made up the training set. Then, 20% of the training data were selected as the validation set, and the rest of the training data were used to train DTLHDM for identifying a safety helmet in the validation data set. After training the DTLHDM, the positive samples from test images and the negative samples from web images with four rounds were verified by the trained DTLHDM to get a report of safety helmet images. The report includes receiver operating characteristic (ROC) curves and precision-recall (PR) curves to show the DTLHDM performance and the comparison of the safety helmet wearing detector with other methods.

3.1 | The proposed FHRM

The FHRM is a safety helmet detector that uses a face detection–based method to get the bounding box of a safety helmet, and it includes a face detector DSFD (J. Li et al., 2019) and a regressor to get the bounding box of the safety helmet. The DSFD extends VGG16 and selects conv3_3, conv4_3, conv5_3, conv_fc7, conv6_2, and conv7_2 to generate original features including of_1, of_2, of_3, of_4, of_5, and of_6 shown in Figure 3. Furthermore, it introduces a FEM (feature enhance module) that makes use of a top–down architecture to fuse information in different layers and learn more context and semantic information from network width and depth, and so the FEM transfers the original features into enhanced features including ef_1, ef_2, ef_3, ef_4, ef_5, and ef_6. In addition, DSFD designs PAL that is based on the differences between low-level and high-level output layers to fit the anchors of various scales, and it makes the model form more effective supervision. Meanwhile, DSFD proposes an anchor matching strategy to improve the match between an anchor and ground truth face and lead to better training results. Therefore, DSFD achieves state-of-the-art detection results on face detection data set and benchmark (FDDB) and WIDER FACE data sets.

Besides the DSFD, the regressor in the FHRM is responsible for a transformation from a face bounding box "*D*" to a safety helmet bounding box "*R*" shown in Figure 3 since the positions of the two bounding boxes are spatially correlated. Inspired by some advanced methods (Felzenszwalb, Girshick, McAllester, & Ramanan, 2014; Girshick, Donahue, Darrell, & Malik, 2014; He, Zhu, Wang, Savvides, & Zhang, 2019), a new method is proposed based on the regression model, which can generate a helmet bounding box from a detected face bounding box. The DSFD first detects the face bounding box "*D*," and then the FHRM computes the helmet bounding box "*R*" using the face bounding box based on our regressor. The face bounding box "*D*" is a helmet region proposal, and the objective of the regressor is that the helmet bounding box "*R*" from the regression method is as close as possible to the helmet bounding box "*G*" based on ground truth. In general, the higher the intersection-over-union (IoU) between the two bounding boxes is, the higher the recognition accuracy is. Since the regression model is trained by the face

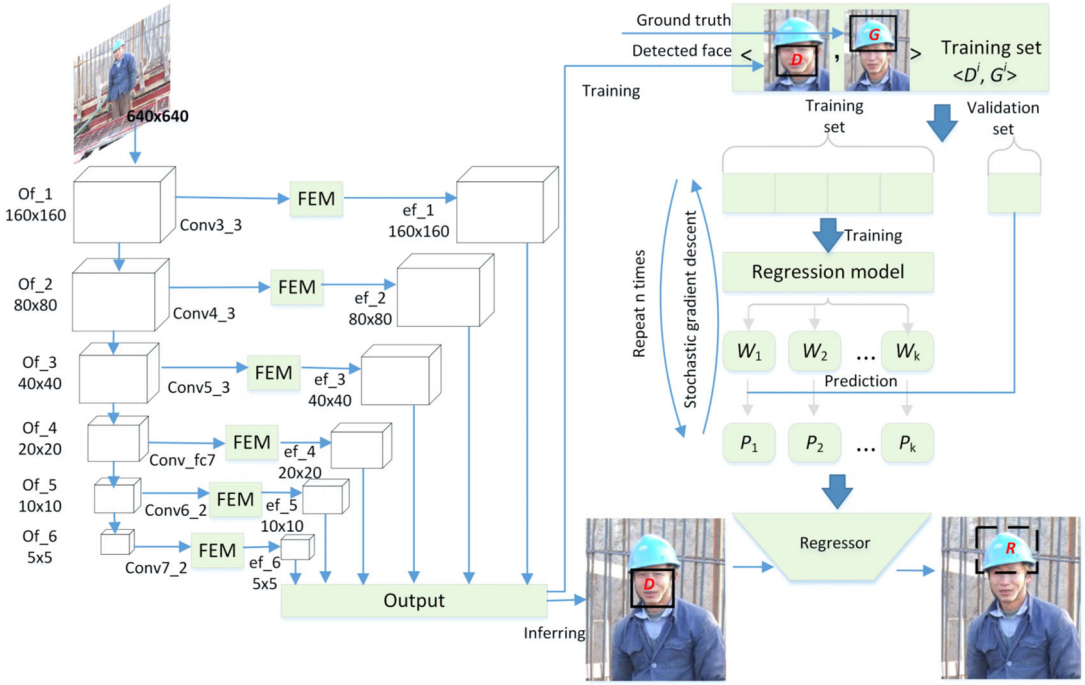


FIGURE 3 The structure of the proposed FHRM

bounding box that is detected by a face detection method, they are closely related. An efficient face detection algorithm not only overcomes the challenging issues for face detection but also helps the regressor get an efficient helmet bounding box with high IoU.

To train the regressor, a set of N training pairs $\{(D^i, G^i)\}_{i=1,\ldots,N}$ is selected, where $D^i = (D^i_x, D^i_y, D^i_w, D^i_h)$ is a vector representing the center coordinates (D^i_x, D^i_y) of a face bounding box D^i together with its width and height (D^i_w, D^i_h). The face bounding box D^i is a helmet region proposal detected by a face detection algorithm. $G^i = (G^i_x, G^i_y, G^i_w, G^i_h)$ is the ground-truth bounding box of the corresponding helmet. Our goal is to learn a transformation function $T(D)$ that transforms a face region proposal D to the ground-truth bounding box G of the corresponding helmet.

We formalize the transform function $T(D)$ according to four functions $T_x(D), T_y(D), T_w(D)$, and $T_h(D)$. Since a neural network can directly learn the scale-invariant feature of an object from a training set, we specify a scale-invariant translation for the first two functions. They indicate the relationship between the center coordinates of the face bounding box D and the regression bounding box R. Here, we assume that there is a similar scale-invariant property between them so that we can map the center of input face proposal D into a predicted helmet bounding box R by the following transformation:

$$R_x = D_w T_x(D) + D_x \tag{1}$$

$$R_y = D_h T_y(D) + D_y \tag{2}$$

Meanwhile, to make the transformations of the width and height for the second two functions have a similar contribution on loss function, we specify the log-space translations for the width and height of bounding box R in the following:

$$\log(R_w) = \log(D_w) + T_w(D) \tag{3}$$

$$\log(R_h) = \log(D_h) + T_h(D) \tag{4}$$

Each function $T_k(D)$ (k is one of x, y, h, w) is modeled by linear regression method using the features of helmet region proposal D including D_x, D_y, D_w, and D_h, which is a vector denoted by $\boldsymbol{F}(D)$. As a result, we have $\boldsymbol{T}_k(D) = \mathbf{v}_k^T \boldsymbol{F}(D)$, where $\mathbf{v}_k^T$ is a vector representing model parameters learned from training samples. We can compute $\hat{\mathbf{v}}_k$ by minimizing the following objective function:

$$\mathbf{v}_k = \underset{\hat{\mathbf{v}}_k}{\operatorname{argmin}} \frac{1}{N} \sum_i^N \left(g_k^i - \hat{\mathbf{v}}_k^T \boldsymbol{F}\left(D^i\right)\right)^2 \tag{5}$$

The g_k includes g_x, g_y, g_h, and g_w , and they are the regression targets for all training pairs $\{(D^i, G^i)\}_{i=1,\ldots,N}$, and are defined as

$$g_x = \frac{(G_x - D_x)}{D_w} \tag{6}$$

$$g_y = \frac{(G_y - D_y)}{D_h} \tag{7}$$

$$g_w = \log\left(\frac{G_w}{D_w}\right) \tag{8}$$

$$g_h = \log\left(\frac{G_h}{D_h}\right) \tag{9}$$

As a linear regression problem, we learn the weights $\hat{\mathbf{v}}_k$ by gradient descent rule to optimize the objective function in Equation (5).

After training, we can compute the regression-based parameters g'_k according to the following equation:

$$g'_k = \hat{\mathbf{v}}_k^T F(D)\, k = x, y, h, w \tag{10}$$

Then, the helmet bounding box is predicted by the regressor using the following equations:

$$R_x = {D_w}^* g'_x + D_x \tag{11}$$

$$R_y = {D_h}^* g'_y + D_y \tag{12}$$

$$R_w = {D_w}^* \exp\left(g'_w\right) \tag{13}$$

$$R_h = {D_h}^* \exp\left(g'_h\right) \tag{14}$$

where $\boldsymbol{R} = (R_x,\ R_y,\ R_W, R_h)$ is a vector representing the center coordinates $(R_x,\ R_y)$ of a helmet bounding box $\boldsymbol{R}$ together with its width and height $(R_w,\ R_h)$.

3.2 | The proposed DTLHDM

The DTLHDM is a safety helmet recognizer to judge whether there is a safety helmet in the helmet bounding box $\boldsymbol{R}$ acquired by the above regression method. The DTLHDM in Figure 4 extends the DenseNet (Huang et al., 2017) network, which is a powerful object feature extractor for object classification problems (Rafiei, & Adeli, 2017b). The DTLHDM is composed of network N_1 and network N_2, and they have the same network structure based on DenseNet. To fine-tune network parameters for safety helmet recognition, we changed the 1,000D fully connected neurons to 2D fully connected neurons for classifying helmet and nonhelmet. The Dense Blocks and convolutional network in network N_1 and network N_2 extract object features, and the fully connected neurons understand these features from a global perspective. And then, using a normalized exponential function softmax is to normalize the global feature into a probability distribution, and so the outputs can be interpreted as probabilities. The larger output corresponds to a larger probability.

Besides changing the DenseNet's network, the pretrained parameters in DenseNet were fine-tuned by helmet training samples to make the DenseNet more suitable for extracting the helmet features.

Meanwhile, we made use of a deep mutual learning method (Y. Zhang, Xiang, Hospedales, & Lu, 2018) to extend the DenseNet's network structure and improve the recognition rate further, which belongs to deep transfer learning since a network learns the knowledge from another network.

Given M samples $\{x_i\}_{i=1}^M$ from two classes including helmet and nonhelmet and the corresponding labels $\{y_i\}_{i=1}^M$ with $y_i \in \{1, 2\}$. For the two-class network, the cross-entropy function is used as the objective function to train network N_1 and N_2. For all samples, we predict the probability of P_i and $(1 - P)_i$, and we define its cross-entropy loss as

$$L_1 = -\left[\sum_{i=1}^{M}\left[y_i \cdot \log\left(P_i\right) + \left(1 - y_i\right) \cdot \log\left(1 - P_i\right)\right]\right] \tag{15}$$

The traditional supervised learning is to train the network and make the inferring results as close as possible to the correct labels for all training samples. To improve the inferring performance of network N_1, we make use of deep mutual learning that is based on its peer network N_2 to provide more training knowledge for its posterior probability Q. Kullback–Leibler *(KL)* divergence is a distance measure of the asymmetry of the difference between two probability distributions, P and Q, which can be defined as follows:

$$KL(Q|P) = \sum_{i=1}^{M}\sum_{j=1}^{2} Q^j\left(x_i\right) \ln\frac{Q^j(x_i)}{P^j(x_i)} \tag{16}$$

The new loss function L_{N_1} and L_{N_2} for network N_1 and N_2 are denoted as

$$L_{N_1} = L_1 + KL(Q|P) \tag{17}$$

$$L_{N_2} = L_2 + KL(P|Q) \tag{18}$$

3.3 | Evaluation metric

The widely used ROC curve and PR (Precision × Recall) curve are adopted to assess the performance of various models. The area under curve (*AUC*) represents the area under an ROC curve, and its value is between 0.5 to 1 where 0.5 means a bad classifier and 1 means an excellent classifier. Furthermore, an ROC space is defined by false positive rate (*FPR*) and true positive rate (*TPR*) as x- and y-axes, respectively. *FPR* defined in (19) indicates the probability that the negative samples are incorrectly predicted as safety helmets by the trained classifier, and *TPR* defined in (20) represents the probability that the positive samples are identified as safety helmets by the trained classifier. In addition, the accuracy (*ACC*) defined in (21) reflects the accuracy of the correct prediction of the trained classifier.

$$FPR = \frac{FP}{FP + TN} \tag{19}$$

$$TPR = \frac{TP}{TP + FN} \tag{20}$$

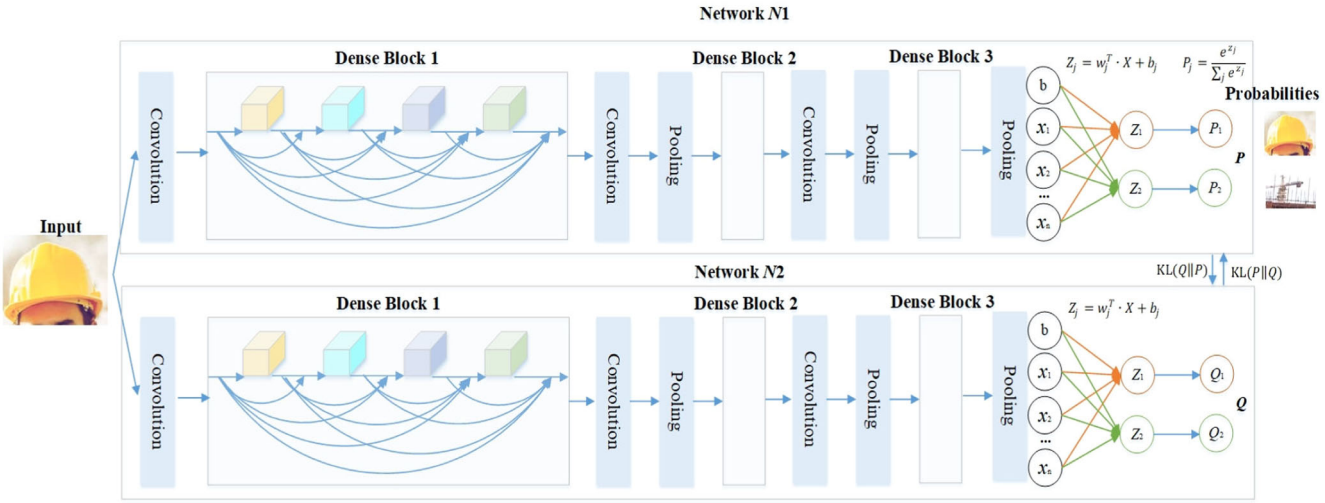


FIGURE 4 The network structure of the proposed DTLHDM

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{21}$$

The *Recall* equals *TPR*, and *Precision* for the PR curve is defined in Equation (22)

$$Precision = \frac{TP}{TP + FP} \tag{22}$$

The true positive (*TP*) is the number of predicted safety helmets that are actually safety helmets. The false negative (*FN*) is the number of predicted nonsafety helmets that are actually safety helmets. The false positive (*FP*) is the number of predicted safety helmets that are actually nonsafety helmets. The true negative (*TN*) is the number of predicted nonsafety helmets that are actually nonsafety helmets.

4 | EXPERIMENT OF FHRM

4.1 | Regressor analysis

The above regressor uses a linear regression method, and the precondition is that there is a linear correlation relationship between the location of the detected face and the location of the corresponding safety helmet. To verify the relationship, we randomly selected 1,000 pairs of (D^i, G^i) to examine the correlation relationships among two variables using the scatterplots in Figure 5, and these plots superimpose smoothed and linear fit lines, which indicates a linear correlation relationship between D and G, especially between the corresponding variables such as D_x and G_x.

Furthermore, we use multiple linear regression to explore the relationship between G_k (k is one of x, y, h, w) and characteristics of the detected face, including D_x, D_y, D_w, and D_h. Then, f-test, t-test, and R^2 correlation test are used to test these regression functions. The f-test indicates overall significance in regression analysis, and the t-test describes the significance of each independent variable. The two tests are usually judged by p-value. If the p-value is less than .01, it shows a significant relationship between G_k (k is one of x, y, h, w) and characteristics of the detected face. The R^2 correlation test is to judge the degree of fitting the regression equation, and its value is between 0 and 1. The closer to 1, the better the degree of fit. Aimed at the above 1,000 data pairs, we constructed the multiple linear regression to verify the relationship between G_k (k is one of x, y, h, w) and characteristics of the detected face. Table 1 shows the p-value and R^2 value. The p-value and R^2 value in Table 1 indicate that the assumption about the linear correlation relationship between the face bounding box D and the corresponding helmet bounding box G is feasible.

TABLE 1 Results for the tests

	p-Value	Multiple R^2	Adjusted R^2
G_x	<2.2e–16	0.9947	0.9947
G_y	<2.2e–16	0.9217	0.9214
G_w	<2.2e–16	0.7939	0.7931
G_h	<2.2e–16	0.704	0.7028

4.2 | FHRM performance

To demonstrate our FHRM model for safety helmet detection, we use the IoU indicator. The IoU can check whether the IoU of safety helmets between the regression box and the ground truth box is greater than a given threshold. If no, the helmet will be identified as "missed," otherwise it will be identified as "successfully detected." Usually, the predicted bounding box of safety helmet with ≥ 0.5 IoU overlap with its ground truth box is regarded as "successfully detected," and the 0.5-IoU has become the de facto metric for object detection problems (He et al., 2019; Zou, Shi, Guo, & Ye, 2019).

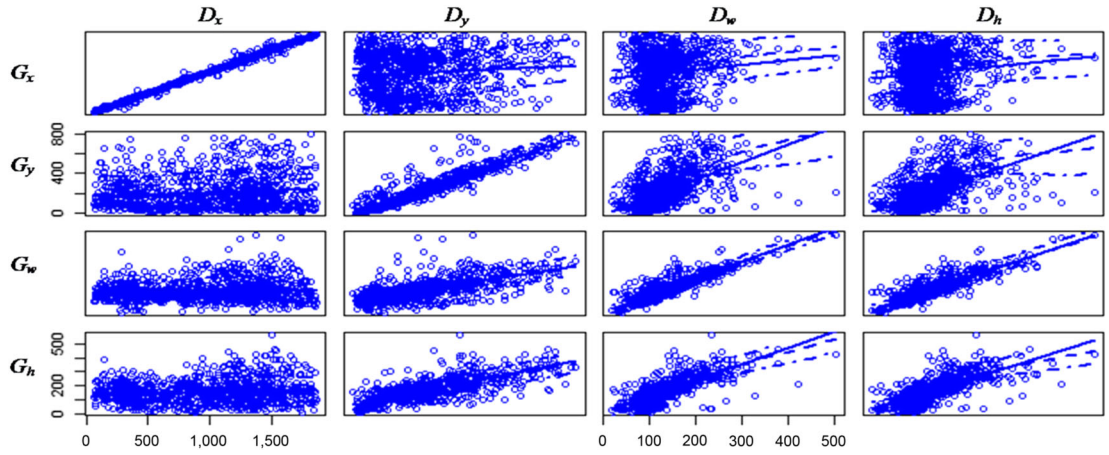


FIGURE 5 Scatterplot matrix

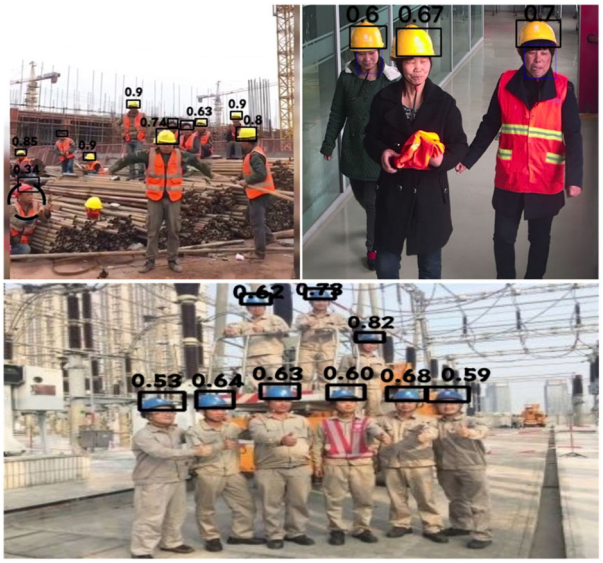


FIGURE 6 The tested images for IoU

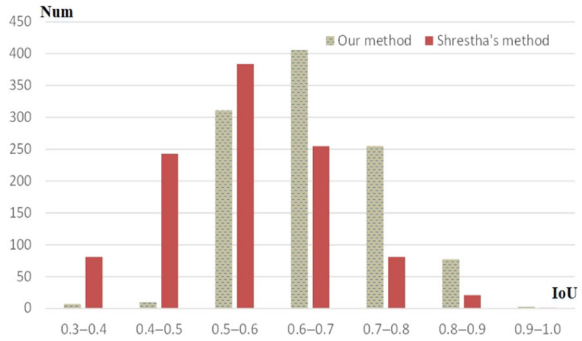


FIGURE 7 The statistic result for IoU

Based on the IoU indicator, 1,069 faces with safety helmets were sampled randomly from the public data set and our private data set. All faces were detected by the proposed FHRM model in Figure 3, and the bounding boxes of safety helmets were acquired by our regression model and the estimation-based method (Shrestha et al., 2015). We computed all IoU values for these safety helmets. Some of the tested images with various challenging issues are chosen and visualized in Figure 6, which shows the regression-based bounding boxes of safety helmet and their IoU values. Meanwhile, Figure 7 illustrates the statistical result of IoU values for the bounding boxes of safety helmets. On the one hand, the percentage of the bounding boxes that are successfully detected by our model is 98.41%, and it is 69.69% by Shrestha's method. The comparison result indicates that the proposed FHRM model is an accurate model to compute the bounding box of safety helmet, and it can provide a good proposal region for safety helmet detection. On the other hand, the proposal region for the bounding box of a safety helmet is identified as "missed" since its IoU value is only 0.34 in Figure 6 (row 1 and column 1) with the dashed circle. The small IoU value usually means that there exists a large angle of head rotation and irregular helmet wearing simultaneously.

5 | EXPERIMENT OF DTLHDM

This section discusses the implementation process to generate the image data set and train the DTLHDM network model for the network parameters. Usually, configuring hyperparameters and training a network is tedious and time-consuming, and we have to explore the experiment by trial-and-error procedure and check all errors. Meanwhile, we chose some advanced methods such as dropout, adaptive learning rate, regularization, and early stopping to improve detection accuracy (Smith & Topin, 2018). All experiments are conducted on a workstation with an NVIDIA RTX 2080Ti 11 GB GPU.

5.1 | Training DTLHDM model

The image data set for training is from the public data set, and these images in the data set were taken on various

TABLE 2 Algorithm for training DTLHDM model

Algorithm: Training Algorithm with deep transfer learning

Input: $\{x_i\}_{i=1}^{M}$: Safety helmet training set;
$\{y_i\}_{i=1}^{M}$: Safety helmet label set;
Learning rate: γ_1 and γ_2;
Initialize: Model parameters v_1 and v_2 for Network N_1 and N_2
Repeat:
$n = n + 1$
Randomly sample training data x_i
Update the outputs P and Q of x_i by normalization
Calculate the gradient descent value and update model parameters v_i (v_1 and v_2):
$v_i \leftarrow v_i + \gamma_i \frac{\partial L(v_i)}{\partial v_i}$
Until: Meet convergence condition

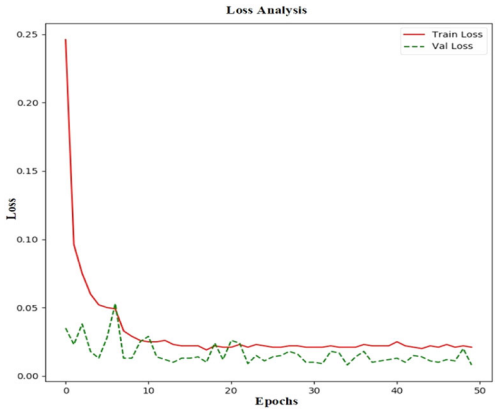


FIGURE 8 Loss for each epoch

construction sites. The number of images is 18,800 including 9,800 for training and 9,000 for testing. It contains 13,963 positive samples and 13,598 negative samples in the training set. Moreover, we separate 20% randomly selected images of training data as a validation set, and 80% images as a training set. Meanwhile, the image data of the four groups are randomly selected from the testing set, and each group contains 1,000 positive samples and 500 negative samples. It is because workers who do not wear safety helmets are a minority in a real scenario.

To train the DTLHDM model in Figure 4, we loaded the weight parameters of pretrained DenseNet and initialized the parameters of the last layer in the model according to the DenseNet initialization method. Table 2 summarizes the training process of the DTLHDM model based on deep transfer learning. We used stochastic-gradient-descent (SGD) with Nesterov momentum, and the momentum was assigned to 0.9. The initial learning rate was assigned to 0.01, which is multiplied by 0.9 every seven epochs. The batch size was set to 4. Considering the DenseNet designed for ImageNet, we resized the input image size to 224 × 224, augmented the training images by a horizontal flip and scale method, and used ImageNet's mean and standard deviation to normalize the image.

Figure 8 shows the loss curves while training the DTLHDM model. With the increase of epoch, the loss curve on the training set decreases rapidly at the beginning and gradually converges. The highest accuracies with loss metric in training and validation are 0.025 at the 43rd epoch and 0.016 at the 33rd epoch, respectively. The GPU boosts the training speed by about 240 min until the 50th epoch, but the training time on the CPU is about 1–2 days. The trained DTLHDM model of the 43rd epoch is chosen for testing, which is detailed in Section 5.2.

5.2 | Ablation study for DTLHDM model

We assess the contribution of deep transfer learning in our DTLHDM model with fine-tuning and mutual distillation methods, and we conduct experiments to compare the feature maps in the first and second layers of the DTLHDM model with DenseNet model. The detailed results with one positive sample and one negative sample are shown in Figure 9. Each image for feature map includes 64 feature maps, and most of the feature maps of the DTLHDM model such as the features in the first two feature maps of the first layer in Figure 9 (row 1) are clearer than the DenseNet and include more features such as the 22nd feature map of the second layer in Figure 9 (row 1). The visualization results demonstrate that discriminately extracting features between the two models help the DTLHDM model boost the detection performance effectively.

To further assess the two models quantitatively, we utilize variance and entropy to measure the feature strength for different models. Variance threshold is a baseline approach to feature selection (Henni, Mezghani, & Gouin-Vallerand, 2018; Q. Zhang, Wang, & Yoon, 2017), and so variance can be used as a measure of feature strength. Meanwhile, Image entropy is an indicator to describe the amount of information, and high entropy image means to have very much contrast and content. The variance and entropy values computed from all feature maps in Figure 9 can distinguish the feature strength.

Figure 10 shows the variance and entropy maps for different feature maps in the first and second layers of rows 1 and 2, and it shows that most of the variance and entropy values for all feature maps computed from our DTLHDM model are more than the values from DenseNet. Moreover, the average values DTLHDM_Avg of variance and entropy values from the DTLHDM model are 0.54 and 0.11 times higher than the average values DenseNet_Avg of variance and entropy values from the DenseNet model in the first layer, and 1.82 and 0.09 times in the second layer. Therefore, our DTLHDM model can extract more features that improve the detection performance of a safety helmet.

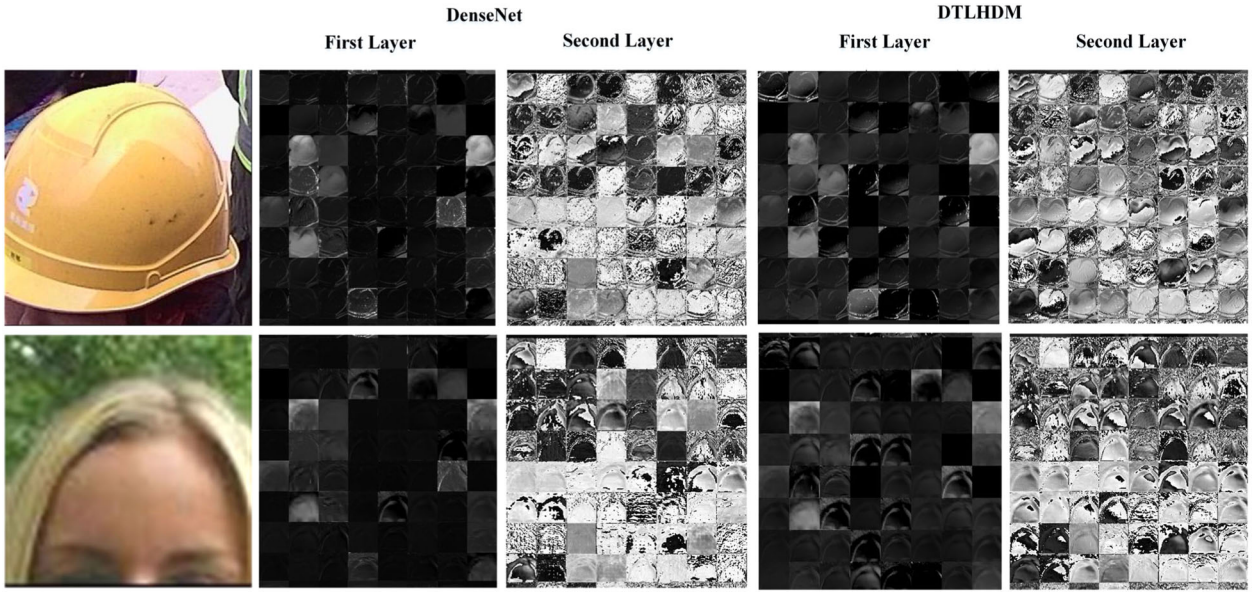


FIGURE 9 The comparison of feature maps between the two models

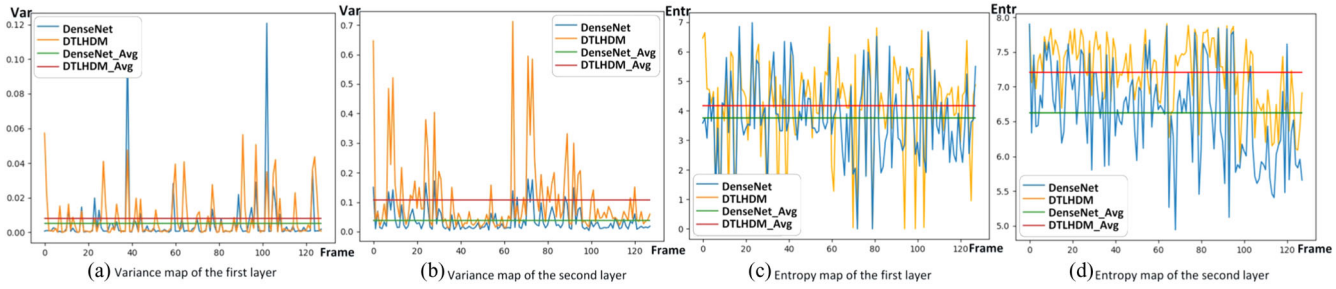


FIGURE 10 Variance and entropy maps for different feature maps

6 | TESTING MODEL AND COMPARATIVE STUDIES

6.1 | Testing the trained model

Based on the above research, 200 original images with 1,996 positive samples and 70 negative samples in the private data set were used to verify the trained FHRM model and the DTLHDM model together. These images with different resolutions that were taken under various conditions cover nearly all types of issues for safety helmet detection including interocclusion, scale variance, perspective distortion, and small object detection. Among them, the number of tiny objects (area $\leq 32^2$ pixels) accounts for 21.4%, and 35.1% for detection objects with occlusion. Some of the tested images with 540 × 720 resolution are chosen and visualized in Figure 11. The original images with detected face bounding boxes and the probability of the corresponding face are in the second row, and the original images with detected safety helmet bounding boxes and the detection results of safety helmet classifier such as 1-0 representing a true positive sample detected as a negative sample are in the third row. The images in the first row are a set of enlarged images.

First, the testing results of face bounding boxes in the second row indicate that the face detection algorithm DSFD is robust to the issues of interocclusion, scale variance, perspective distortion, facial expression, face pose, and small face. In our experiment, the DSFD algorithm has detected all faces that human eyes can normally identify, especially the small face with occlusion in the blue dashed circle of the first image and the whole side face in the blue dashed circle of the first image of the second row. Second, the bounding boxes of safety helmets in Figure 11 show that our FHRM model can compute them accurately. Moreover, it can achieve our regressor goal to some challenging face poses such as the faces in the blue dashed circles of the fourth and fifth images of the second row.

Third, there are three misjudged samples with the 1-0 missing alarm for safety helmet recognition, and it means that a safety helmet is detected as a nonsafety-helmet-use sample.



FIGURE 11 Visualization of detected faces and helmets

The reason for the misjudgment of safety helmets in the blue dashed circles of the first and third images of the third row should be due to small objects, looking-up, and cross-scene. Another reason for the misjudgment of safety helmet in the blue dashed circle of the second image of the first row is the overlapping between safety helmets and occlusion issue that leads to incomplete features. In addition, there is one misjudged sample with 0–1 false alarm in the blue dashed circle of the third image of the third row, the reason for misjudgment should be that the safety helmet is relatively small and the background contains some features such as edge features that are similar to the features of a safety helmet. Finally, although there exist various challenging issues on construction sites, our model achieves 94.68% in *ACC* for the testing data set.

6.2 | Comparative studies

A wide range of collected images from the public testing data set was tested by the DTLHDM model, and it provides a confidence value for detected safety helmets. The confidence value is regarded as the probability of being a safety helmet. For example, a 0.8 confidence value indicates that the probability of the safety helmet in a bounding box is 80%. The safety helmet is identified when the confidence value is above the confidence threshold, which determines the classification of helmets and nonhelmets. Using different confidence thresholds is to classify positive and negative samples and form a PR curve, and the PR curve for evaluating the DTLHDM model is shown in Figure 12a. To compare our model with other models, Figures 12b and 12c show the Faster R-CNN-based PR curve (Q. Fang et al., 2018), ViBe+C4+CFD (color feature discrimination) and ViBe+HOG+SVM+CFD-based PR curves (K. Li et al., 2018), respectively. The three diagonal lines in Figure 12 indicate that the average precision (Average precision (AP) is the area under the PR curve) value of our model is the highest and up to 98.53%.

TABLE 3 The comparison of the two models

Method	*Precision (%)*	*Recall (%)*
Visual range	95.9	94.5
Weather	95.2	94.6
Illumination	96.5	94.9
Posture	95.2	95.7
Occlusion	94.4	87.4
Average (Q. Fang et al., 2018)	95.4	93.4
Our model	96.2	96.2

Furthermore, a high confidence threshold is to reject more ambiguous safety helmet samples and produce high precision and missing rate. To have a win–win result between precision and recall, we selected the C point of PR curve in Figure 12a as the confidence threshold for comparative studies, and its value for *Precision* and *Recall* is 0.962. In Table 3, we compared our model with the state-of-the-art method based on deep learning (Q. Fang et al., 2018), which analyzed the impacts of visual range, weather, illumination, posture, and occlusion on *Precision* and *Recall* indicators. Here, we took

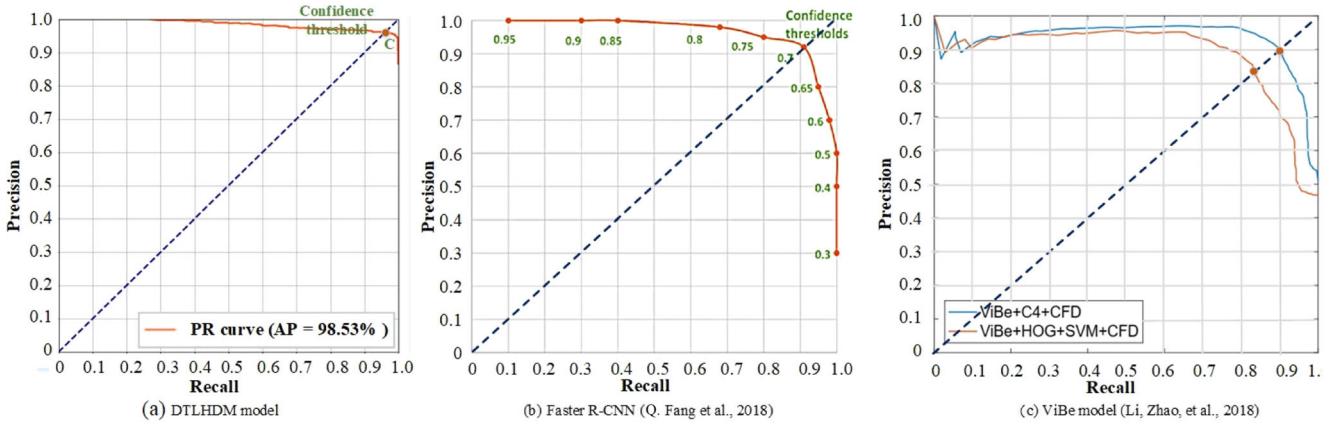


FIGURE 12 PR curves for different models

their average to compare the two models. The *Precision* of our model is higher than the Faster R-CNN model (Q. Fang et al., 2018) by 0.8% (96.2% vs. 95.4%), and the *Recall* by 2.8% (96.2% vs. 93.4%). Table 3 shows that the Faster R-CNN model suffers from occlusion issue since it requires detecting the whole workers with NHU. However, our model using the face region of a worker makes the testing results less sensitive to occlusion issues, and Figure 1 illustrates the comparison. Moreover, the Faster R-CNN model (Q. Fang et al., 2018) depends on the bounding box of the whole worker. The bounding box inevitably contains many background pixels that possibly take up nearly half of the box area, and it results in uncertainty and ambiguity for object detector.

Aside from the PR curve analysis, the ROC curve is meaningful in evaluating the performance of the proposed model. Figure 13 shows the ROC curves of different models. The *AUC* indicator is used to evaluate the quality of different models. The *AUC* of the HOG and SVM-based method is 89.20%, and the *AUC* of the C4 and CFD-based model (K. Li et al., 2018) is 94.13%. The *AUC* of our model is 97.64%, and it is improved by 8.44% and 3.51%, respectively.

In addition, we assessed the stability of our model using the *ACC* indicator. Four independent groups of data were sampled randomly from the testing set. The *ACC* values of four groups of data in Figure 14 are 93.6%, 95.0%, 94.87%, and 94.4%, respectively. The average *ACC* of correct prediction on trained safety helmet classifier reaches 94.47%. Therefore, our model is generally stable.

Finally, to further compare our model with Faster R-CNN-based method (Q. Fang et al., 2018), we randomly chose some image frames from our private testing data set, which are used to test the impact of occlusion issues. We classify the occlusion degree into two categories: "face visible" and "only part of the face visible" as shown in Table 4. The testing results indicate that the precision is improved by 7%, and the recall for "only part of face visible" issue is done by 6.1%. The results for the Faster R-CNN in Table 4 are better than the

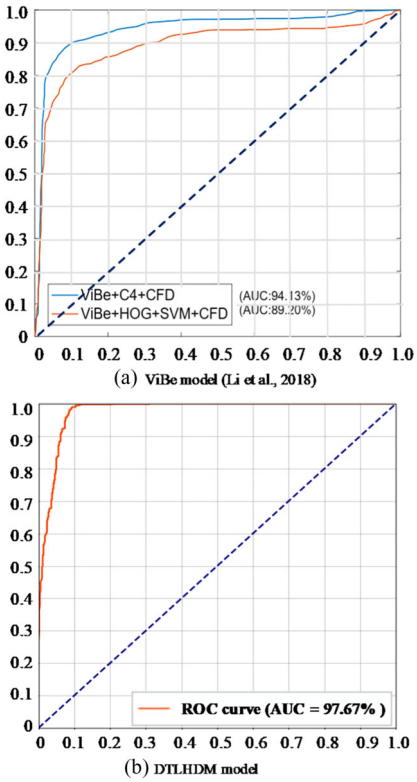


FIGURE 13 The ROC curves for different models

Faster R-CNN adopted by Q. Fang et al. (2018). On the one hand, the testing data set is different. On the other hand, the newest Faster R-CNN is used from MMDetection (Chen et al., 2019), which achieves state-of-the-art results for pedestrian detection. All in all, our method has better performance for occlusion problems.

6.3 | Model analysis

To make the model more transparent and understand how to make decisions from a deep network model, we used the

TABLE 4 Precision and recall under different occlusions

Categories	Method	TP	FN	FP	*Precision* (%)	*Recall* (%)
Face visible	Faster R-CNN	734	30	63	92.1	96.1
	Our model	756	8	7	99.1	99.0
Only part of face visible	Faster R-CNN	247	83	25	90.8	74.8
	Our model	267	63	6	97.8	80.9

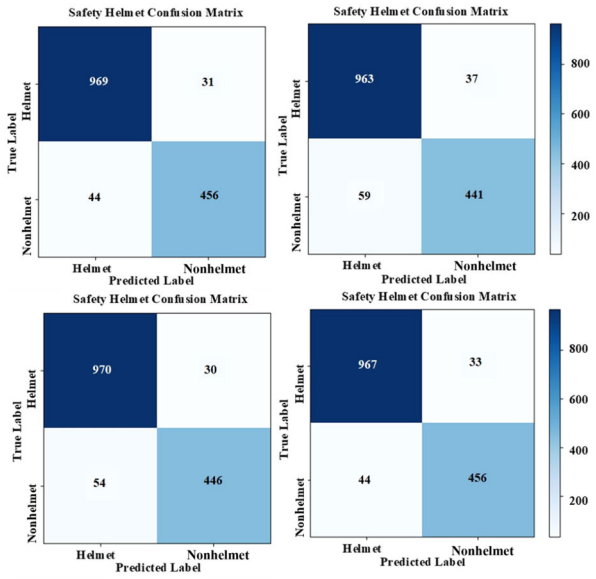


FIGURE 14 The confusion matrices for ACC

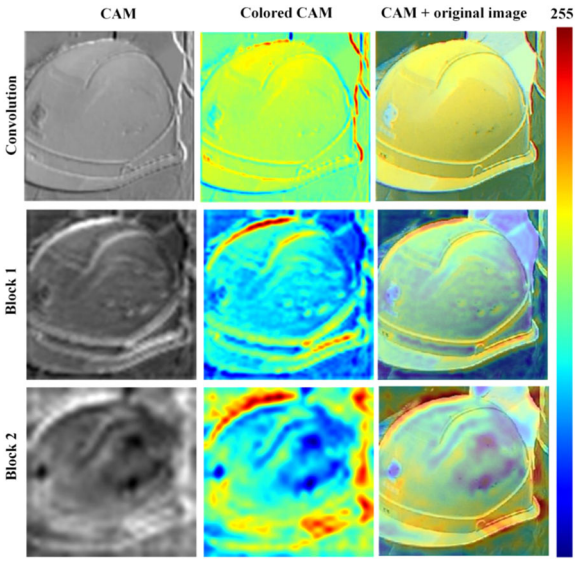


FIGURE 15 Visualization results using CAM

class activation mapping (CAM) method (Zhou, Khosla, Lapedriza, Oliva, & Torralba, 2016) to produce visual explanations in Figure 15. The CAM maps the predicted class results back to the previous CNN layer to generate a visual activation map for highlighting the important regions in an input image. Figure 15 shows the CAM maps of a safety helmet image, the colored CAM maps based on the right color table, and the original images with the colored CAM maps, and the output is mapped back to the block 2, block 1, and convolution layer in Figure 4 to produce the CAM maps. These maps visualize the output on the image to highlight the discriminative helmet parts detected by our model. We found that the model indeed concentrates on the safety helmet in Figure 15 (column 1) and its edges in Figure 15 (columns 2 and 3). These visualization results help us establish an appropriate trust in detection tasks from our model and discern a strong network model.

7 | CONCLUSIONS

A deep learning-based method for safety helmet detection on various construction sites was presented using face bounding-box regression and deep transfer learning methods. The performance of the trained model was evaluated on the public and private data sets. The test samples were input into the model, and the face bounding boxes, safety helmet bounding boxes, and the results of safety helmet detection were consequently obtained. Although test samples contained various challenging issues with interocclusion, scale variances, perspective distortion, and small objects, the test results for safety helmet detection on all images presented the consistent performance and good stability for our model. Experimental results indicate that our model achieves state-of-the-art results compared to the ViBe method (K. Li et al., 2018) and the Faster R-CNN method (Q. Fang et al., 2018). According to Table 3, our model exceeds the Faster R-CNN by 0.8% (96.2% vs. 95.4%) in *Precision* and by 2.8% (96.2% vs. 93.4%) in *Recall*, and the *AP* value of our model is the highest in all models. Moreover, the *AUC* of our model for the ROC curve is 97.64%, and it is improved by 3.54% (97.67% vs. 94.13%). The proposed method has an average running time of 362 ms in all test samples. The average processing time for DSFD is 170 ms per frame, and it is 16 ms per frame for DTLHDM. Since there are no generally and publicly accepted data sets for evaluating safety helmet wearing detection models under various scenes until now, we did not compare the time performance with other existing methods for safety helmet detection.

The main contributions of this work are as follows: (a) The methodological contribution is to propose a new method and idea for detecting safety helmet wearing in various construction sites where interocclusion, scale variances, perspective distortion, and small objects occurred. These issues are challenging for all detection objects. However, face detection is one of the most successful areas of artificial intelligence research, and this area is to accumulate various challenging data sets for training an advanced face detection model. We made use of a regression method with a state-of-the-art face detection model to overcome these issues. Meanwhile, the face detection–based method can help to solve the issue of the carrier recognition of a safety helmet since the face is regarded as an efficient and legal biometric feature to authenticate a person. (b) We propose a face bounding-box regression model FHRM with the detection of the face of various sizes to determine the candidate region of a safety helmet. This method has achieved an efficient IoU to help a detector detect a safety helmet accurately. Furthermore, the method is very different from the Faster R-CNN method (Q. Fang et al., 2018) since it is less sensitive to occlusion issues. (c) We make use of fine-tuning and mutual distillation methods for improving safety helmet recognition based on DenseNet (Huang et al., 2017), and the ability of the DTLHDM model to extract features shown in Figure 10 is better than the pretrained DenseNet. In addition, the face detection–based method makes safety helmet recognition less sensitive to background pixels compared to the pedestrian-based method (Q. Fang et al., 2018), and its accuracy outperforms previous methods. (d) We present extensive analysis and experiments on the public data set and verify state-of-the-art performance, and it is possible for the public data set to become a benchmark dataset to compare different models.

Although our model achieved promising results, the proposed method has several limitations. First, our model has no way to detect the workers with their back to the surveillance camera. Indeed, the face detection–based method has many advantages such as more data sets, better models, and fewer sensitivities to background noise. Moreover, compared to the pedestrian-based method (Q. Fang et al., 2018), the method can detect safety helmets and identify their colors that are important for construction management to signify different roles on construction sites. The face-based method also helps a management system recognize the carrier of safety helmet and provides efficient credentials. Therefore, it is worth considering and deploying onsite surveillance cameras properly to overcome the disadvantages of the face-based method. Second, our model needs to add a face recognition algorithm to identify the carrier of a safety helmet. Since there are many excellent algorithms such as ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019) and Sphereface (Liu et al., 2017) in the field, they were not mentioned in this article. Third, a good regression model of a safety helmet bounding box is essential for improving identification accuracy. Currently, the predicted bounding box of safety helmet with $\geq$0.5 IoU overlap with its ground truth box is regarded as "successfully detected," and so it is possible to improve the IoU threshold and successfully detected rate by a better regression model. Moreover, current data sets such as WIDER FACE data set (Yang, Luo, Loy, & Tang, 2016) and MS-COCO data set (Lin et al., 2014) that is one of the most popular data sets for object detection rarely include some challenging issues such as rain, smoke, and dust weather, and so further enhancements in the model will improve its performance to deal with various challenging issues. Finally, a generally and publicly accepted data set for evaluating different models of safety helmet wearing detection is crucial, and it should cover various challenges and different construction sites. The results of comparison and testing on such a data set will be more reasonable.

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